

Mahavir Polytechnic, Nashik
Department of Computer Engineering

Year: TY

Subject: STE (315323)

UNIT 5: Software Project Management

Marks: 8

COURSE LEVEL LEARNING OUTCOME:

CO5	Apply project management techniques in software development
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THEORY LEARNING OUTCOMES (TLO'S) ALIGNED TO COS:

TLO 5.1	Apply CPM/PERT scheduling technique for software project.
TLO 5.2	Construct timeline chart/ Gantt chart to track progress of the given software project.

LEARNING CONTENT MAPPED WITH THEORY LEARNING OUTCOMES (TLO'S) AND COS:

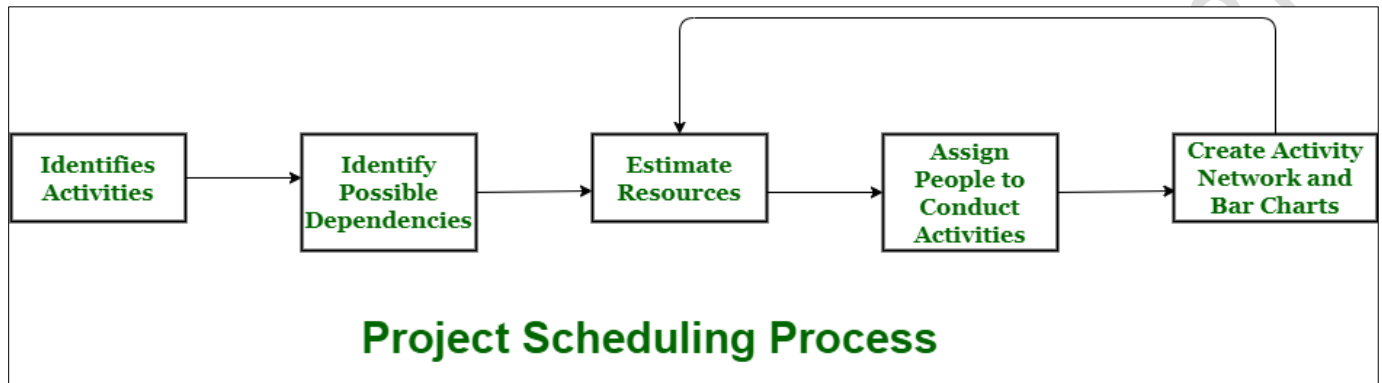
TLO 5.1	Project Scheduling: Basic principles, Work breakdown structure, Activity network
TLO 5.2	Project Tracking: Timeline charts, Earned Value Analysis, Gantt Charts
TLO 5.3	Scheduling techniques: Critical Path Method(CPM), Program Evaluation Review Technique(PERT)

➤ TLO 5.1

➤ Project Scheduling:

- Project scheduling in software engineering is the systematic process of organizing and sequencing project tasks, defining dependencies, estimating durations, allocating resources, and assigning timelines to achieve project goals within a specified timeframe.

- It involves creating a detailed roadmap, often visualized in tools like [Gantt charts](#), to guide the development process, ensuring smooth execution, timely completion, and increased customer satisfaction.
- Key steps include defining activities, creating a Work Breakdown Structure (WBS), estimating time and resources, identifying task dependencies, determining the critical path, and using project scheduling software for planning and tracking.



5.1.1 Basic principles: The basic principles of software project scheduling include identifying project goals, breaking the project into smaller tasks, determining task dependencies, estimating resource needs and durations, allocating resources, identifying the critical path, and monitoring progress against the schedule using tools like Gantt charts.

➤ Core principles

- **Compartmentalization:** A project is broken down into smaller, distinct, and manageable tasks. This is often done using a Work Breakdown Structure (WBS), which provides a hierarchical view of the project's activities.
- **Interdependency:** The relationships between tasks are identified, showing which must be completed in sequence and which can be done in parallel. A key activity is determining the **critical path**—the longest sequence of dependent tasks that defines the project's total duration.
- **Time allocation:** Realistic start and end dates are assigned to each task based on its estimated effort and dependencies. This includes factoring in buffer time for unexpected delays.
- **Effort validation:** Before assigning tasks, resource estimates are validated to ensure the right personnel, skills, and tools are available when needed. It is important to avoid over-allocating resources.
- **Defined responsibilities:** Each task is assigned to a specific team member, clarifying who is responsible for each part of the project.
- **Defined outcomes:** Each task must have a clear, defined outcome or deliverable. These outcomes are reviewed for quality and are crucial for marking milestones.
- **Defined milestones:** Significant checkpoints are set throughout the timeline to mark the completion of major phases or deliverables.

➤ Common scheduling techniques

Software projects can use various techniques to create and manage schedules:

- **Critical Path Method (CPM):** Identifies the longest sequence of dependent tasks (the "critical path") to determine the shortest possible project completion time.
- **Program Evaluation and Review Technique (PERT):** Uses a three-point estimation approach—optimistic, pessimistic, and most likely—to account for task duration uncertainty.
- **Gantt charts:** A visual, horizontal bar chart that represents the project timeline, showing start and end dates for each task and their dependencies.
- **Agile scheduling:** Breaks work into small, iterative "sprints" or periods. This adaptive approach is suitable for projects with evolving requirements, as seen in Scrum and Kanban.
- **Resource leveling:** A technique used to balance resource allocation and prevent overworking team members by adjusting the schedule.

Creating a software project schedule (general steps)

While there is no single PDF, the process follows these common steps:

1. **Define project scope:** Clearly define project goals, objectives, and deliverables. This prevents "scope creep," where extra features are added late in the process.
2. **Identify tasks:** Break down the project into a Work Breakdown Structure (WBS) that lists all activities and subtasks.
3. **Establish dependencies:** Determine the order in which tasks must be completed.
4. **Estimate time and resources:** Estimate the duration of each task and identify the human and non-human resources required.
5. **Visualize the schedule:** Use tools like Gantt charts to create a visual representation of the project timeline.
6. **Review and monitor progress:** Track task completion and compare actual progress against the initial schedule. Adapt the plan as needed.
7. **Communicate:** Regularly communicate progress to all stakeholders, including team members and clients.

5.1.2 Work breakdown structure:

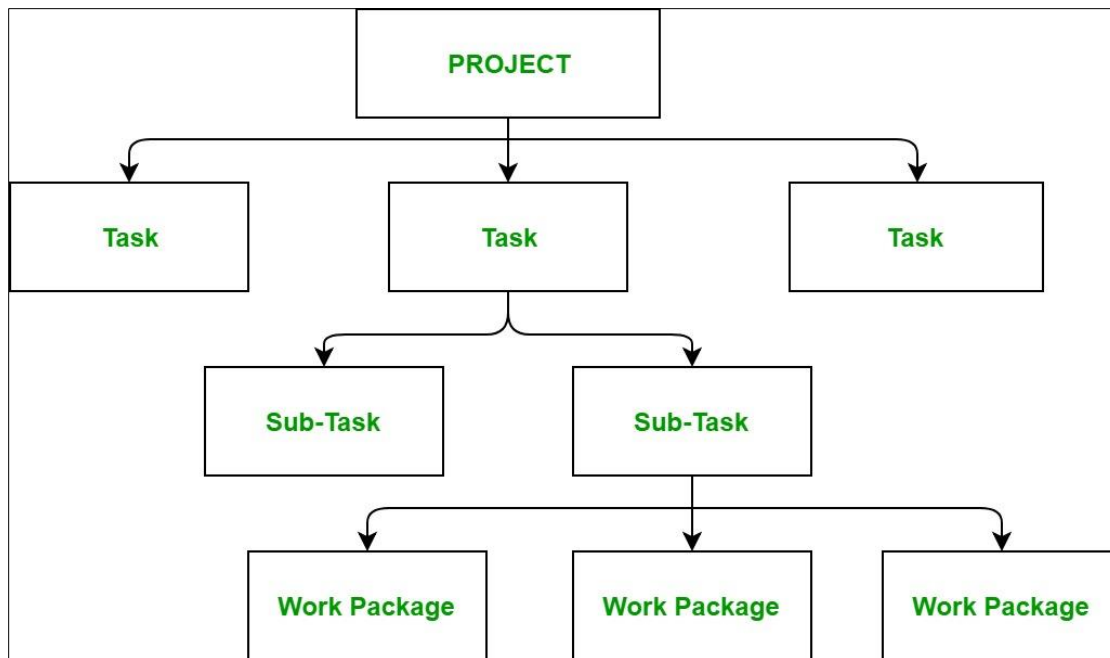
A Work Breakdown Structure includes dividing a large and complex project into simpler, manageable, and independent tasks. The root of this tree (structure) is labeled by the Project name itself. For constructing a work breakdown structure, each node is recursively decomposed into smaller sub-activities, until at the leaf level, the activities become undividable and independent. It follows a Top-Down approach.

Steps Work Breakdown Structure:

Step 1: Identify the major activities of the project.

Step 2: Identify the sub-activities of the major activities.

Step 3: Repeat till undividable, simple, and independent activities are created.



➤ Construction of Work Breakdown Structure

1. Firstly, the project managers and top level management identifies the main deliverables of the project.
2. After this important step, these main deliverables are broke down into smaller higher-level tasks and this complete process is done recursively to produce much smaller independent tasks.
3. It depends on the project manager and team that upto which level of detail they want to break down their project.
4. Generally the lowest level tasks are the most simplest and independent tasks and takes less than two weeks worth of work.
5. Hence, there is no rule for upto which level we may build the work breakdown structure of the project as it totally depends upon the type of project we are working on and the management of the company.
6. The efficiency and success of the whole project majorly depends on the quality of the Work Breakdown Structure of the project and hence, it implies its importance.

➤ Uses of Work Breakdown Structure

1. **Cost estimation:** It allows doing a precise cost estimation of each activity.
2. **Time estimation:** It allows estimating the time that each activity will take more precisely.
3. **Easy project management:** It allows easy management of the project.
4. **Helps in project organization:** It helps in proper organization of the project by the top management.

➤ Example:

A Work Breakdown Structure (WBS) example is a hierarchical decomposition of a project into manageable deliverables and tasks. For instance, a "Building a House" project (Level 1) can be broken into major phases like "Design" and "Utilities" (Level 2), then further into sub-activities such as "Planning and Changes" for Design or "Plumbing" and "Electrical" for Utilities (Level 3).

Here's a WBS example for "Baking a Cake":

- Level 1: Baking a Cake: (The project's overall objective)
 - Level 2: Prepare Batter
 - Level 3: Mix Ingredients:
 - Level 4: Measure Dry Ingredients
 - Level 4: Cream Butter and Sugar
 - Level 4: Add Eggs and Vanilla
 - Level 4: Combine Wet and Dry Ingredients
 - Level 2: Bake the Cake
 - Level 3: Preheat Oven
 - Level 3: Prepare Cake Pan
 - Level 3: Bake until Done
 - Level 3: Cool the Cake
 - Level 2: Decorate the Cake
 - Level 3: Make Frosting
 - Level 3: Frost the Cake
 - Level 3: Add Toppings
-

5.1.3 Activity network:

- An activity network diagram visually represents a project's tasks, their sequence, and dependencies to determine the critical path and overall project duration.
- It uses nodes to represent activities or events and arrows to show their logical flow. This graphical tool helps in identifying bottlenecks, managing resources, and ensuring project timelines are met.

➤ How it works:

1. List Activities:

Break down the project into individual activities.

2. Define Dependencies:

Determine the logical relationships and order in which activities must occur.

3. Represent Activities:

- **Activity-on-Node (AON):** This modern method uses nodes (often boxes) to represent activities and arrows to show the dependencies between them.
- **Activity-on-Arrow (AOA):** In this older method, arrows represent the activities, and the nodes represent events or milestones.

4. Visualize Flow:

The diagram is drawn from left to right to reflect the project's timeline, showing which activities can run in parallel and which must be done in series.

5. Identify the Critical Path:

This is the longest sequence of tasks that determines the shortest possible time to complete the project. Delays on the critical path directly delay the entire project.

6. Determine Float:

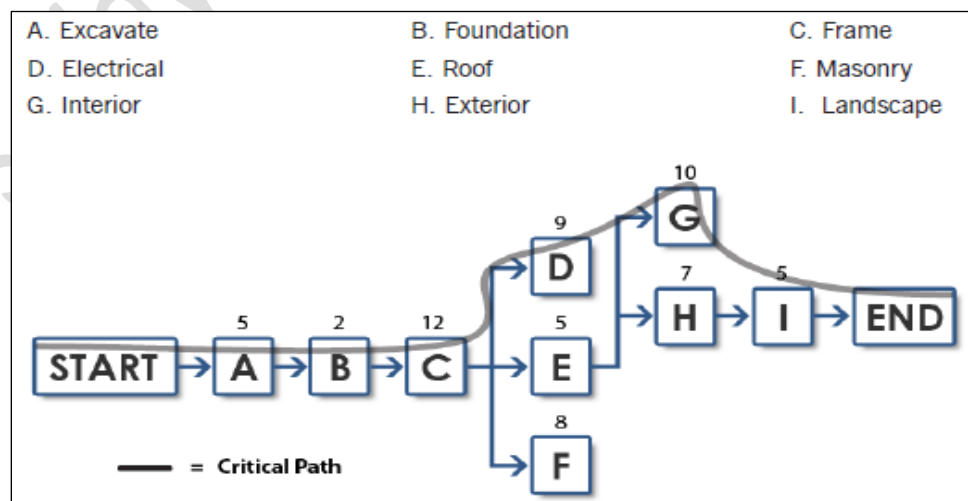
The diagram helps identify "float" or "slack," which is the amount of time an activity can be delayed without impacting the project's overall end date.

- **Benefits in Project Scheduling**

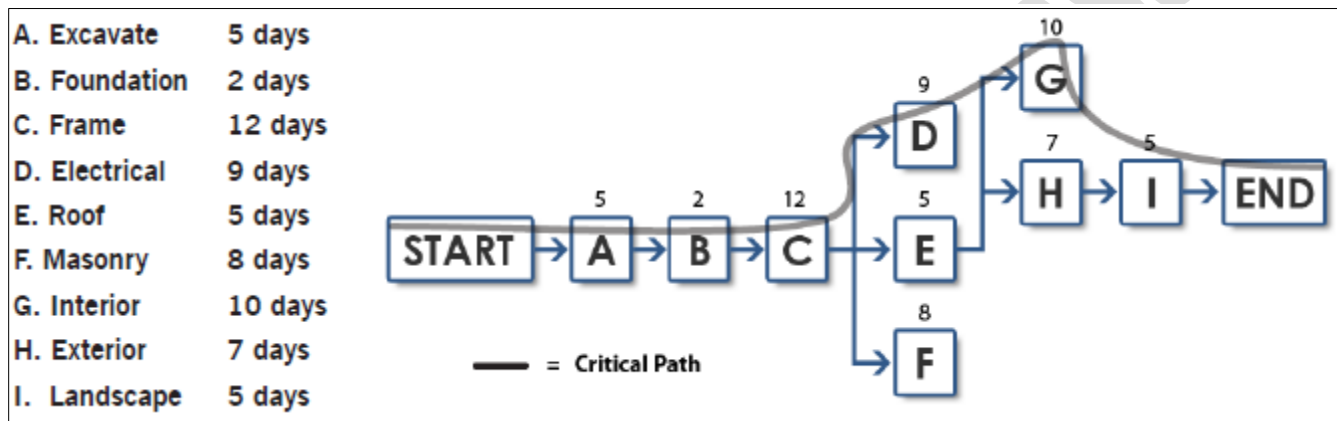
- **Visualizes Dependencies:** Clearly illustrates how tasks are interconnected, helping to understand the project's structure.
- **Determines Project Duration:** Calculates the total time needed for the project.
- **Identifies Bottlenecks:** Highlights potential roadblocks or areas of inefficiency.
- **Resource Allocation:** Helps in understanding resource needs and allocating them effectively.
- **Improved Communication:** Provides a clear, easy-to-understand overview for project teams and stakeholders.

- An **Activity Network Diagram** is a diagram of project activities that shows the sequential relationships of activities using arrows and nodes.
- An activity network diagram tool is used extensively in and is necessary for the identification of a project's critical path (which is used to determine the expected completion time of the project).

Example: Suppose the team is tasked with improving the process of building a house. The team lists the major steps involved – everything from the excavation step through the landscaping step.



- The team creates a chart – **Activity Network Diagram** – where the nodes (the boxes) represent the nine major steps involved in building a house. Arrows that connect the nodes show the flow of the process.
- Some of the process steps (nodes A, B, and C) run in **series**, while other process steps (nodes D, E, and F) run in **parallel**. Notice that Step B cannot happen until step A has been completed. Likewise, step C cannot happen until step B has completed. Step H cannot happen until steps D, E, and F have completed – and ALL need to be completed before Step H.
- So, nodes A, B, and C are running in series. Nodes D, E, and F run in parallel. This is important to know because those steps that are running in parallel most likely will have different expected completion times.



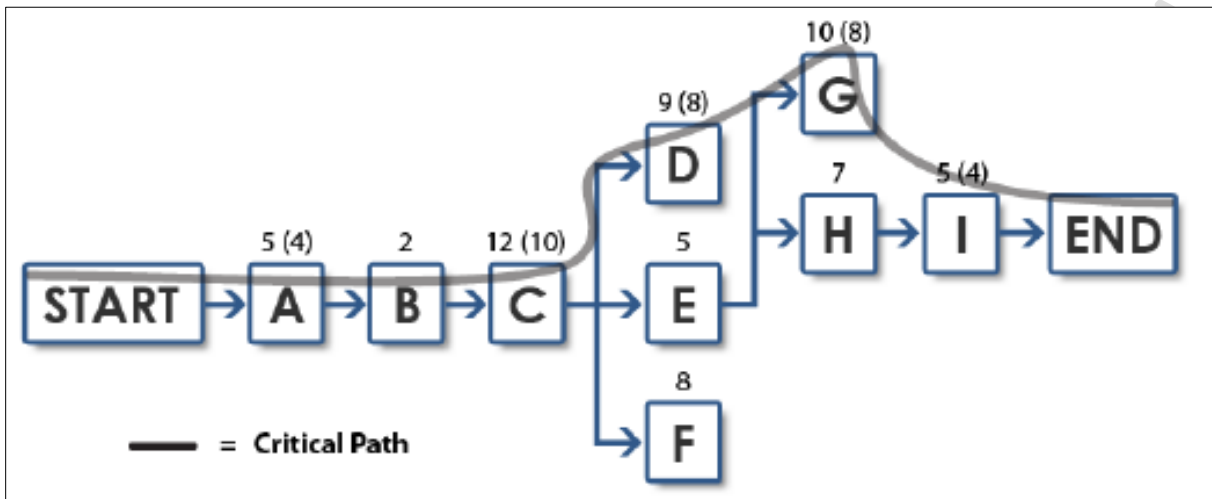
- Critical Path:**

The team's job is to take note of which of the nodes D, E, and F, will be taking the most amount of time, and which of those nodes is expected to take the least amount of time. This is essential when creating the **Critical Path**. For instance, if node D is expected to take the most amount of time as compared with nodes E and F, it is not important that nodes D and E start at the exact same time as node F. Those steps can start later, but they have to be finished no later than the most time consuming of the three steps that run in parallel. The team evaluates the nine steps and come to a consensus on how many days each of the nine steps will take. The critical path is a line that goes through all of the nodes that have the longest expected completion times.

- Most Likely Time:**

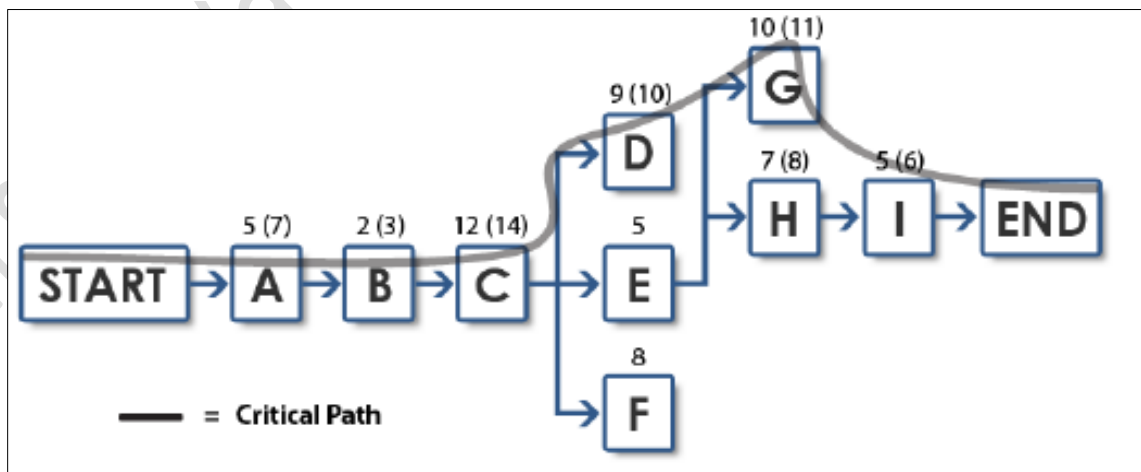
Nodes A, B, and C run in series, so the critical path is straightforward. Notice that between the three nodes that run in parallel, (nodes D, E, and F) node D is expected to take the longest to complete as compared to the other two nodes. The critical path would run through nodes D and G because those particular nodes have the longest expected completion times. The line above shows the critical path. By looking at the Activity Network Diagram the team can easily see that the expected completion time as defined by the critical path is 50 days. $(5+2+12+9+10+7+5 = 50 \text{ days})$ That's the **MOST LIKELY** time.

- **Optimistic Time:**
- The team might want to know what the best case (**Optimistic Time**), in terms of time, would be. To come up with that number, the team would decide upon the shortest possible time for each of the nodes, and then add those up. The numbers in parenthesis are the most optimistic times.
 $(4+2+10+8+8+7+4 = 43)$



Pessimistic Time:

- The team also might want to know what the worst case (**Pessimistic Time**), in terms of time, would be. To come up with that number, the team would decide upon the longest possible time for each of the nodes, and then add those up.
- Note: To determine the best case or the worst case, the critical path line must be followed. The numbers in parentheses are the most pessimistic times. $(7+3+14+10+11+8+6 = 59)$ Remember, you are only calculating the numbers along the critical path when calculating the most optimistic and pessimistic times.



Expected Time

So what does all of this mean? It means the project most likely will take 50 days, but it could take 59 days, or it can be done as soon as 43 days.

$$\begin{aligned}\text{Expected Time} &= \frac{\text{Optimistic} + [4 (\text{Most Likely})] + \text{Pessimistic}}{6} = \\ \text{Expected Time} &= \frac{43 + 200 + 59}{6} = 50.3 \text{ days}\end{aligned}$$

Control Bands

We could calculate control bands around the average. Here's how we do that:

$$\begin{aligned}\text{Limits of expected variation} &= \frac{\text{Optimistic} - \text{Pessimistic}}{6} = \\ \text{Limits of expected variation} &= \frac{59 - 43}{6} = \\ \text{Limits of expected variation} &= \frac{16}{6} = 2.7\end{aligned}$$

For the critical path, we can expect the project to take from 47.6 days to 53.0 days

$50.3 + 2.7 = 53$ on the high side

$50.3 - 2.7 = 47.6$ on the low side.

5.2 Project Tracking:

Project scheduling involves creating a detailed plan that maps out all the tasks, their sequences, durations, and resource assignments, with the goal of setting a timeline for project completion. Project tracking is the ongoing process of monitoring progress against this schedule, identifying deviations, and taking corrective action to ensure the project stays on course, meets its deadlines, and stays within budget.

➤ **Techniques/Tools:**

- **Gantt Charts:** Visual representations of the project timeline, showing tasks as bars along a timeline.
- **Critical Path Method (CPM):** A technique for identifying the longest sequence of dependent tasks (the critical path), which determines the shortest possible project duration.

- **Program Evaluation and Review Technique (PERT)**: A method for estimating task durations and identifying the critical path, particularly useful for projects with high uncertainty.
- **Task Lists**: A straightforward method for listing tasks and their sequences, best for simpler projects.

➤ **Project Tracking (The Monitoring)**

Tracking is the dynamic process of monitoring the execution of the project against the established schedule.

- **Purpose:**

To monitor progress, identify potential issues or deviations from the plan, and ensure the project stays on track.

➤ **Key Activities:**

- **Progress Monitoring**: Continuously collecting and evaluating data on completed tasks, achieved milestones, and resource utilization.
- **Performance Measurement**: Using metrics to assess how well the project is performing against its planned schedule and budget.
- **Issue Identification**: Proactively spotting bottlenecks, risks, and other problems before they significantly impact the project.
- **Corrective Action**: Implementing timely interventions and making necessary adjustments to the plan to mitigate risks and resolve issues.
- **Reporting**: Providing real-time information and insights to stakeholders and senior management.

➤ **Tools & Techniques:**

- **Project Management Software**: Tools that offer dashboards, Gantt charts, and reporting features for real-time progress updates.
- **Time-Tracking Software**: Features that allow team members to log hours spent on specific tasks.
- **Regular Meetings**: Team meetings to discuss progress, challenges, and upcoming activities.

5.2.1 Timeline charts:

In software engineering, a timeline chart (most often a [Gantt chart](#)) is a project management tool that visualizes project tasks, milestones, and their dependencies over a specific time period. It uses horizontal bars to represent tasks, showing their start and end dates, duration, and overall progress. Timeline charts are essential for tracking software development progress, managing resources, identifying bottlenecks, and communicating the project schedule to stakeholders.

➤ **Key Components of a Timeline Chart**

- **Tasks:** Individual pieces of work within the project, depicted as horizontal bars.
 - **Milestones:** Significant points or events in the project lifecycle, often marked as specific points or flags on the timeline.
 - **Dependencies:** Relationships between tasks, indicating that one task must be completed before another can begin.
 - **Start and End Dates:** The scheduled beginning and completion times for each task.
 - **Progress Tracking:** Visual cues, such as shading within the bar, can show how much of a task has been completed.
 - **Timeline Axis:** A horizontal axis representing time, often showing weeks, months, or quarters.
- How It's Used in Software Engineering

- **Project Planning:**
Outlining all project phases, key activities, and their estimated timelines.
- **Progress Monitoring:**
Providing a clear, at-a-glance overview of project status and task completion.
- **Resource Management:**
Illustrating who is responsible for each task and ensuring that resources are allocated efficiently over time.
- **Identifying Bottlenecks:**
Making it easier to spot potential delays or critical paths in the project schedule.
- **Stakeholder Communication:**
Simplifying complex project schedules for team members, managers, and other stakeholders.

➤ **Types of Timelines**

While Gantt charts are the most common in software, other timeline formats exist:

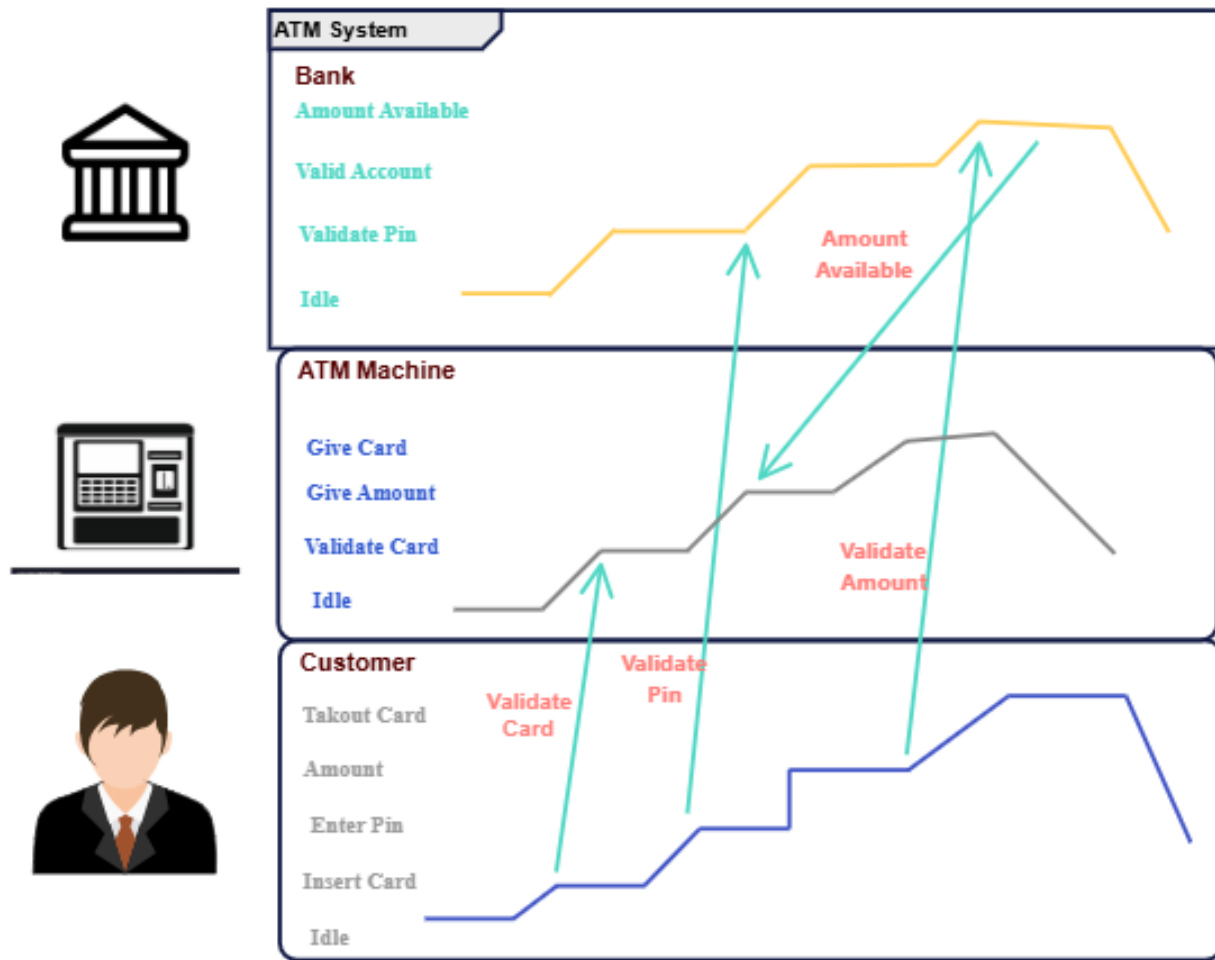
- **Horizontal Timelines:** A chronological sequence of events shown from left to right.
- **Vertical Timelines:** Events displayed from top to bottom.
- **Roadmaps:** High-level overviews of product strategy and development over longer periods.

Example:**Q. Prepare timeline chart for ATM machine.**

A timeline chart for an ATM machine's operation isn't a historical timeline of its invention, but rather a sequence of events in a single transaction, starting with the user inserting a card, authenticating with a PIN, selecting a transaction, entering an amount, and finally receiving cash and a receipt. The ATM communicates with the bank's server at multiple points to validate the card and account balance, ensuring the transaction's security and successful completion.

Here is a step-by-step timeline for a typical ATM cash withdrawal:

1. **Insert Card:** The user inserts their debit or credit card into the ATM.
2. **Card Validation:** The ATM reads the card and sends the information to the bank's server to verify its validity.
3. **PIN Entry:** The bank server confirms the card is valid and sends a message to the ATM. The ATM then prompts the user to enter their PIN.
4. **PIN Validation:** The entered PIN is sent to the bank server for authentication.
5. **Transaction Selection:** If the PIN is valid, the bank server sends a confirmation to the ATM, which then displays the main menu. The user selects their desired transaction type (e.g., withdrawal) and account type.
6. **Withdrawal Amount:** The user enters the desired withdrawal amount.
7. **Fund Check:** The ATM sends the withdrawal request to the bank server to check for sufficient funds in the account.
8. **Dispense Cash:** If funds are available, the bank server approves the transaction, the ATM counts the cash, and then dispenses it to the user.
9. **Receipt & End Transaction:** The ATM prompts the user if they want a receipt. After the user's choice, the ATM ejects the card, and the transaction is complete.



5.2.2 Earned Value Analysis:

- Earned Value Analysis (EVA) is a project management technique that integrates cost, schedule, and scope to measure project performance by tracking the value of work completed against the planned budget and timeline.
- It provides project managers with insights into project health, allowing them to forecast future outcomes, identify risks like cost overruns or schedule delays, and take proactive measures to ensure project success.
- The core of EVA lies in comparing three key metrics: the Planned Value (PV), the value of the work scheduled; the Earned Value (EV), the value of the work actually performed; and the Actual Cost (AC), the real cost incurred for the performed work.

➤ Key Concepts and Metrics

EVA relies on three fundamental quantities to measure performance:

- **Planned Value (PV) / Budgeted Cost of Work Scheduled (BCWS):**

The budgeted cost of all the work scheduled to be completed by a specific point in time.

- **Earned Value (EV) / Budgeted Cost of Work Performed (BCWP):**

The budgeted cost of the work that has actually been completed. This is the core "earned" value.

- **Actual Cost (AC) / Actual Cost of Work Performed (ACWP):**

The actual cost incurred in accomplishing the work performed up to that point.

➤ **How It Works**

1. **Measure Work Performed:**

Instead of just looking at costs or schedules, EVA measures the value of the work that has been accomplished.

2. **Compare Plan vs. Reality:**

By comparing PV, EV, and AC, managers can determine if they are ahead or behind schedule and over or under budget.

3. **Calculate Performance Indicators:**

Using these metrics, project managers calculate performance indices (like the Schedule Performance Index and Cost Performance Index) to understand performance trends.

4. **Forecast Future Performance:**

These trends are then used to forecast the project's future cost and completion date.

➤ **Benefits in Software Engineering**

- **Comprehensive Performance View:**

EVA offers a holistic view by combining cost, schedule, and scope, which is more informative than traditional cost and schedule reports alone.

- **Proactive Issue Identification:**

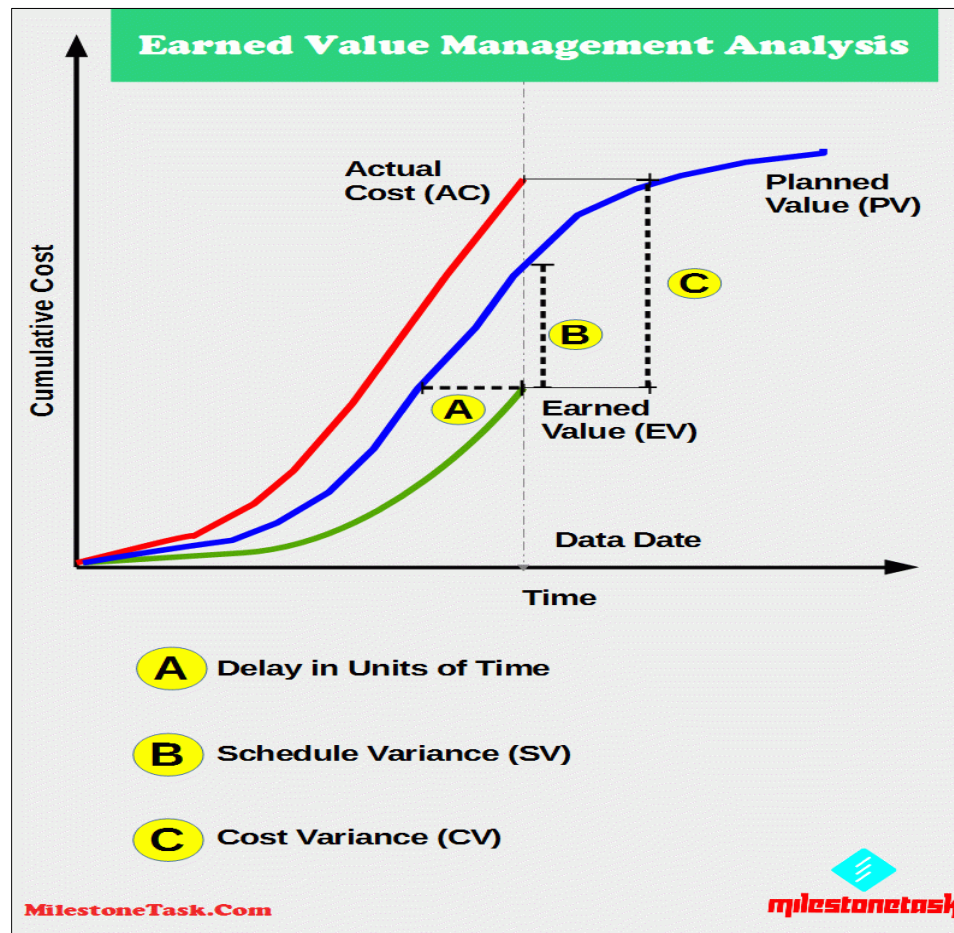
It provides early warnings of potential problems, allowing project teams to address issues before they lead to significant overruns or delays.

- **Informed Decision-Making:**

Managers gain the data needed to make confident decisions about scope, schedule, and resource allocation.

- **Improved Communication:**

EVA provides a common, objective basis for communicating project status and performance to stakeholders.



Example Scenario

Imagine a project to build a house with a total budget of \$500,000.

1. At 50% completion (planned):

- $PV = \$500,000 * 50\% = \$250,000$.

2. Actual Progress & Cost:

- You've actually completed 40% of the work, and it cost you \$220,000.
- $EV = \$500,000 * 40\% = \$200,000$.
- $AC = \$220,000$.

3. Calculations:

- $SV = EV - PV = \$200,000 - \$250,000 = -\$50,000$ (Behind Schedule).
- $CV = EV - AC = \$200,000 - \$220,000 = -\$20,000$ (Over Budget).
- $SPI = EV / PV = \$200,000 / \$250,000 = 0.80$ (Behind Schedule).
- $CPI = EV / AC = \$200,000 / \$220,000 = 0.91$ (Spending inefficiently).

5.2.3 Gantt Charts:

Generalized Activity Normalization Time Table (GANTT) chart is type of chart in which series of horizontal lines are present that show the amount of work done or production completed in given period of time in relation to amount planned for those projects. It is horizontal bar chart developed by Henry L. Gantt (American engineer and social scientist) in 1917 as production control tool. It is simply used for graphical representation of schedule that helps to plan in an efficient way, coordinate, and track some particular tasks in project.

The purpose of Gantt chart is to emphasize scope of individual tasks. Hence set of tasks is given as input to Gantt chart. Gantt chart is also known as timeline chart. It can be developed for entire project or it can be developed for individual functions. In most of projects, after generation of timeline chart, project tables are prepared. In project tables, all tasks are listed in proper manner along with start date and end date and information related to it.

Gantt chart represents following things :

- All the tasks are listed at leftmost column.
- The horizontal bars indicate or represent required time by corresponding particular task.
- When occurring of multiple horizontal bars takes place at same time on calendar, then that means concurrency can be applied for performing particular tasks.
- The diamonds indicate milestones.

Advantages :

- **Simplify Project -**
Gantt charts are generally used for simplifying complex projects.
- **Establish Schedule -**
It simply establishes initial project schedule in which it mentions who is going to do what, when, and how much time it will take to complete it.
- **Provide Efficiency -**
It brings efficiency in planning and allows team to better coordinate project activities.
- **Emphasize on scope -**
It helps in emphasizing i.e., gives importance to scope of individual tasks.
- **Ease at understanding -**
It makes it easy for stakeholders to understand timeline and brings clarity of dates.
- **Visualize project -**
It helps in clearly visualizing project management, project tasks involved.
- **Organize thoughts and Highly visible -**
It organizes your thoughts and can be highly visible so that everyone in enterprises can have basic level of understanding and have knowledge about what's happening in project even if they are not

involved in working.

- **Make Practical and Realistic planning -**

It makes the project planning practical and realistic as realistic planning generally helps to avoid any kind of delays and losses of many that can arise.

Disadvantages :

- Sometimes, using Gantt chart makes project more complex.
- The size of bar chart does not necessarily indicate amount of work done in project.
- Gantt charts and projects are needed to be updated on regular basis.
- It is not possible or difficult to view this chart on one sheet of paper. The software products that produce Gantt chart needed to be viewed on computer screen so that whole project can be seen easily.

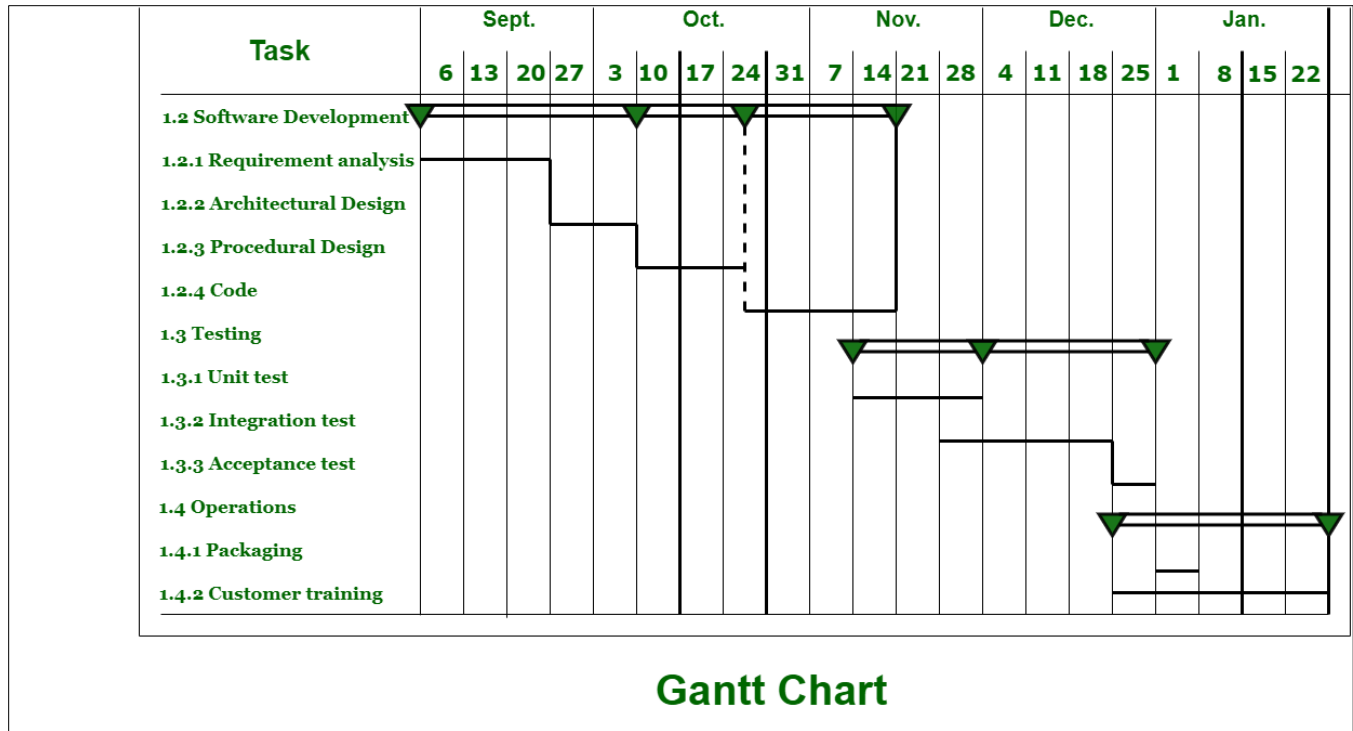
Applications :

There are several professions, where use of gantt chart is very beneficial. Some of them are given below:

- **Advertising Manager -**
Advertising Managers generally controls and supervises end result of advertising companies, scheduling advertisements in different media, etc.
- **Operations Manager -**
Operations Managers generally control and handle resources that are essential for company operations.
- **Project Manager -**
Project Managers generally motivates project teams, collaborate with team members, schedule task and complete that on time, and report to stakeholders, etc.

Example :

Nowadays, there are many companies and teams that use Gantt chart to plan, schedule, and execute their projects. Some of them are consulting agencies, manufacturing companies, Marketing teams, Construction companies, etc. Below is an example of Gantt chart:



➤ **5.3 Scheduling techniques:**

➤ **Core Techniques**

• **Gantt Charts:**

Visual tools that display project tasks and durations on a timeline, providing a clear overview of the project schedule.

• **Critical Path Method (CPM):**

A technique that identifies the longest sequence of dependent tasks in a project, determining the minimum time needed for completion and highlighting critical activities.

• **Program Evaluation and Review Technique (PERT):**

Used for projects with uncertain or variable task durations, PERT estimates project completion time by considering best, worst, and most likely scenarios.

• **Work Breakdown Structure (WBS):**

A hierarchical decomposition of a project into smaller, manageable tasks, which forms the foundation for detailed scheduling.

• **Task Dependencies:**

Identifying the logical sequence of tasks, understanding which tasks must finish before others can start.

- **Milestones:**

Key events or checkpoints in the project timeline that signify the completion of a significant phase or delivery.

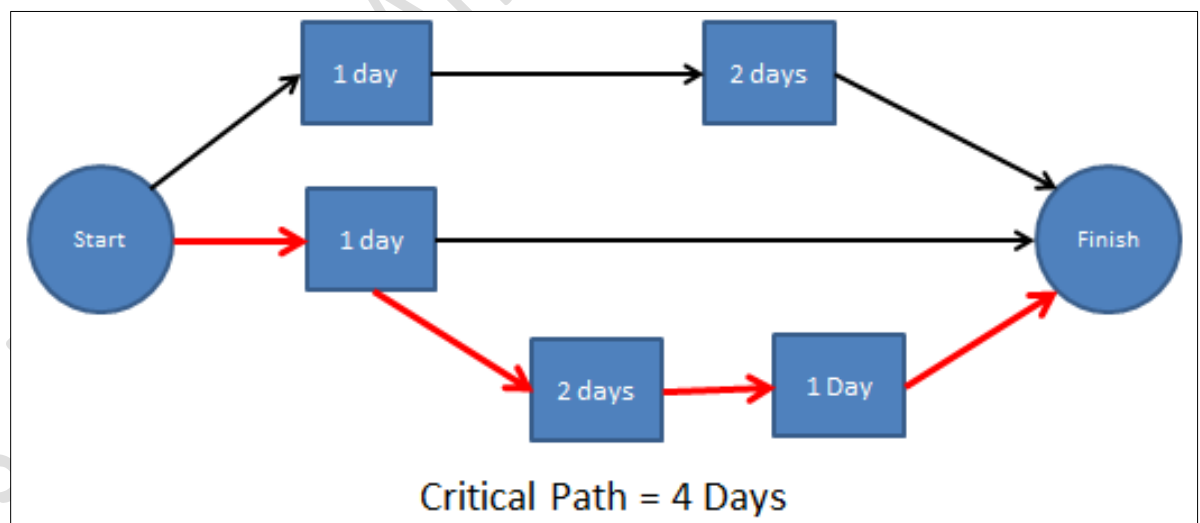
5.3.1 Critical Path Method(CPM):

The Critical Path Method (CPM) is a way to determine the critical path of a schedule. The critical path is the longest path of scheduled activities that must be met to execute a project.

This is important for [Program Managers \(PM\)](#) to know since any problems on the critical path can prevent a project from moving forward and being delayed.

[Earned Value Management \(EVM\)](#) analysis focuses on critical and near-critical paths to identify cost and schedule risks. Unlike the critical path, other schedule paths might have slack time to avoid delaying the entire project. There might be multiple critical paths on a project.

The method is used with [Program Evaluation and Review Techniques \(PERT\)](#).



Critical Path Method (CPM) Development Steps

The seven (7) steps in the CPM are: [1]

- Step 1: List of all activities required to complete the project (see Work Breakdown Structure (WBS)),
- Step 2: Determine the sequence of activities
- Step 3: Draw a network diagram
- Step 4: Determine the time that each activity will take to completion
- Step 5: Determine the dependencies between the activities
- Step 6: Determine the critical path
- Step 7: Update the network diagram as the project progresses

The CPM calculates the longest path of planned activities to the end of the project and the earliest and latest that each activity can start and finish without making the project longer. This process determines which activities are “critical” (i.e., on the longest path) and which have “total float” (i.e., can be delayed without making the project longer).

5.3.2 Program Evaluation Review Technique(PERT):

The Programme Evaluation and Review Technique (PERT) is a project management tool that uses a flowchart or network diagram to break down a project into tasks, estimate task durations, and identify dependencies between tasks.

PERT addresses uncertainty by using three time estimates (optimistic, most likely, and pessimistic) to calculate a weighted average for each task's duration, helping to determine the project's minimum completion time and the critical path.

It was developed by the U.S. Navy in the 1950s for managing complex projects like the Polaris submarine program and remains useful for large-scale, non-routine projects with uncertain timelines.

➤ When to Use PERT

- **Complex, Large-Scale Projects:** Especially those that are one-time or non-routine.
- **Projects with Uncertain Timelines:** When it's difficult to know the precise duration of individual tasks.
- **Research and Development Projects:** Where unpredictability is a significant factor.

How it Works (Simplified)

1. **List Activities:** Identify all the tasks required for the project.
2. **Identify Dependencies:** Determine which tasks depend on others to be completed.
3. **Estimate Time:** For each task, provide optimistic, most likely, and pessimistic time estimates.
4. **Calculate Expected Time:** Use a formula to find the weighted average of the three estimates.

5. **Draw the PERT Chart:** Create a network diagram (arrows and nodes) showing the tasks, events, and their dependencies.
6. **Identify the Critical Path:** Analyze the network to find the longest sequence of activities that dictates the project's minimum duration.

➤ **Apply CPM/PERT scheduling technique for software project.**

To apply PERT/CPM to a software project,

- 1) list all tasks
- 2) define dependencies
- 3) create a network diagram
- 4) estimate durations (using optimistic, most likely, and pessimistic times for PERT, or a single estimate for CPM)
- 5) identify the critical path (the longest sequence of tasks with no float)
- 6) monitor the path to manage timelines, resources, and potential delays. PERT is best for uncertain projects, while CPM is better for projects with predictable task durations, [according to Atlassian](#)

➤ **Steps to Apply PERT/CPM**

1. **Identify Tasks and Dependencies:**

- Break down the software project into all its essential tasks.
- Determine the relationships and sequence between tasks, identifying which must be completed before others can begin.

2. **Create a Network Diagram:**

- Draw a diagram to visualize the project's activities and their dependencies, representing them as nodes or arrows.

3. **Estimate Task Durations:**

- **For PERT:** For each task, provide three time estimates:
- Optimistic time (O): The best-case scenario.
- Most likely time (M): The most probable duration.
- Pessimistic time (P): The worst-case scenario.
- A weighted average $(O + 4M + P) / 6$ is often used to calculate the expected duration.
- **For CPM:** Use a single, predictable time estimate for each task.

4. **Identify the Critical Path:**

- The critical path is the longest sequence of tasks from the start to the end of the project, which determines the shortest possible project duration.

- Any delay on a task along the critical path will directly delay the entire project.

5. **Calculate Float (Slack):**

- For each task, calculate the "float" or "slack," which is the amount of time it can be delayed without delaying the project's completion.

6. **Monitor and Control:**

- Regularly monitor the critical path and the project's progress.
- Use this information to make timely decisions, reallocate resources, manage risks, and ensure the project stays on schedule and within budget.

When to Use PERT vs. CPM

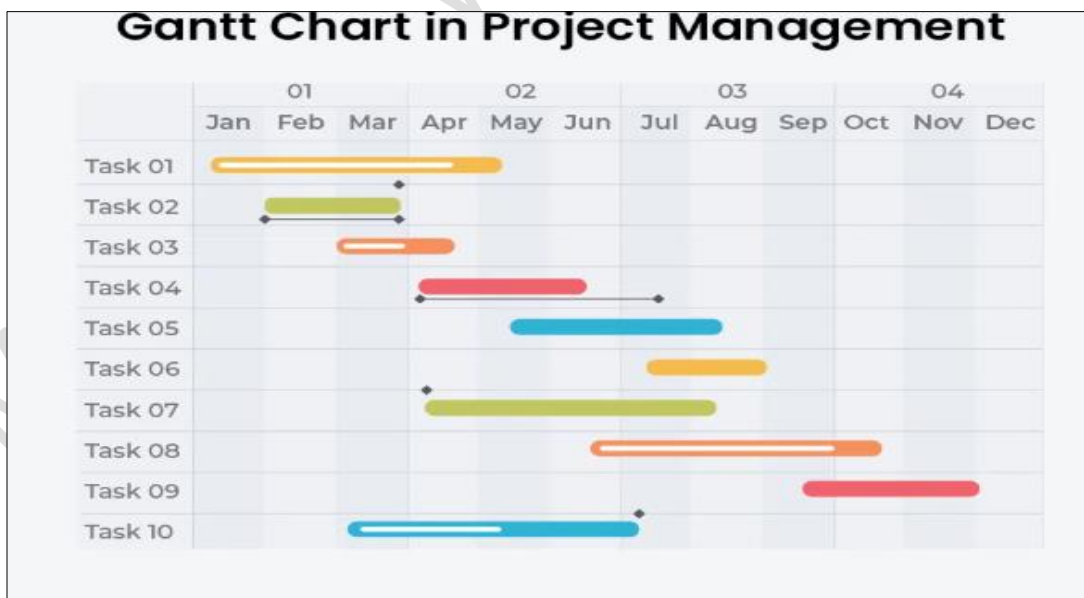
- **PERT:**

Best for projects with high uncertainty, such as new software development or R&D projects, where task durations are difficult to predict.

- **CPM:**

More suitable for projects with well-defined activities and predictable task durations, allowing for efficient resource allocation and focus on meeting deadlines.

Construct timeline chart/ Gantt chart to track progress of the given software project.



Important Questions:

2 Marks Questions:**1. Define PERT and CPM.(Refer 5.3.2)**

Ans:

PERT:

The Programme Evaluation and Review Technique (PERT) is a project management tool that uses a flowchart or network diagram to break down a project into tasks, estimate task durations, and identify dependencies between tasks.

CPM:

A technique that identifies the longest sequence of dependent tasks in a project, determining the minimum time needed for completion and highlighting critical activities.

2. Define Project Scheduling.

Ans:

Project scheduling in software engineering is the systematic process of organizing and sequencing project tasks, defining dependencies, estimating durations, allocating resources, and assigning timelines to achieve project goals within a specified timeframe.

3. What is Gantt Chart? State its Advantages.

Ans:

Visual tools that display project tasks and durations on a timeline, providing a clear overview of the project schedule. Following are the advantage of Gantt Chart.

- 1. Easy identification of tasks:** The Gantt Chart helps us to easily identify the various tasks, their inter-relation, and the Progress of the tasks.
- 2. Helps to evaluate critical resources utilization:** We can evaluate the critical Resources and their utilization, Dependencies, and Impacts of schedule changes in the Gantt Chart. Along with this, we can also create Visual alerts in case of scheduling conflicts.
- 3. Helps to manage tasks order:** The Gantt chart equips the project experts with the capabilities to manage the order in which tasks need to be carried out so that we can easily identify the priority of the activities.

Note: for 2 mark question only list advantages, no need to explain. For 4 marks question explain advantages in detail.

4 Marks Questions:

1. Explain Work breakdown Structure with suitable diagram.(5.1.2)
2. State and Explain any 4 Principles of project scheduling.(5.1.1)
3. Prepare timeline chart for ATM machine.(5.2.1)
4. Prepare macro time line chart for 15 days of college management system (5 days a week).

Ans:

A macro timeline for a 15-day, 5-days-a-week college management system project, when considering SDLC phases, would allocate roughly 1-2 days per week to Requirements Gathering, Analysis, and

Design, 1-2 days per week to Development and Testing, and 1 day per week to Deployment and Maintenance, with the remaining days for initial planning and finalization. This high-level view helps in understanding the broad project flow over the 15-day period.

Macro Time Line (15 Days / 3 Weeks)

This timeline assumes a 5-day work week (Monday to Friday).

- **Week 1: Foundation & Planning**
- **Day 1-2 (Monday-Tuesday):** Project Initiation & Requirements Gathering. Kick-off meeting, defining the project scope and high-level requirements for the college management system.
- **Day 3-4 (Wednesday-Thursday):** Requirements Analysis & High-Level Design. Analyzing the gathered requirements and sketching a basic, high-level architecture for the system.
- **Day 5 (Friday):** Detailed Design & Planning. Begin creating detailed design specifications for key modules and finalize a detailed project plan for the subsequent weeks.
- **Week 2: Development & Initial Testing**
- **Day 6-7 (Monday-Tuesday):** Module Development (Phase 1). Developers start building critical modules, such as student registration and course management.
- **Day 8-9 (Wednesday-Thursday):** Module Development (Phase 2) & Unit Testing. Continue developing other modules, like faculty profiles and attendance tracking, and begin unit testing completed modules.
- **Day 10 (Friday):** Integration Testing (Initial). Integrate the developed modules and conduct initial integration testing to find and fix early integration issues.
- **Week 3: Testing, Deployment & Wrap-up**
- **Day 11-12 (Monday-Tuesday):** System Testing & User Acceptance Testing (UAT) Preparation. Conduct comprehensive system testing for the integrated system and prepare for user acceptance testing.
- **Day 13 (Wednesday):** User Acceptance Testing (UAT). Key stakeholders or end-users test the system to ensure it meets their needs and requirements.
- **Day 14 (Thursday):** Deployment & Training (Initial). Deploy the approved system and conduct initial training for administrators and key users.
- **Day 15 (Friday):** Final Review & Handover. Conduct a final project review, address any immediate post-deployment issues, and formally hand over the system.

5. Explain GANTT chart for project tracking with an example. (5.2.3)

6. Prepare Gantt chart for hostel management system (Five days a week) consider phases of SDLC.

Ans:

A Gantt chart for a hostel management system's SDLC (Software Development Life Cycle) would map tasks across the phases of Planning, Requirements Analysis, Design, Development (Coding), Testing, Deployment, and Maintenance, with tasks assigned to a five-day work week. The chart would show task durations, dependencies, and milestones, enabling project management and resource allocation.

1. Planning Phase

- **Task:** Project scoping, feasibility study, and resource allocation.
- **Duration:** ~1-2 weeks.
- **Dependencies:** N/A.

2. Requirements Analysis Phase

- **Task:** Gather detailed requirements from stakeholders, document functional and non-functional requirements.

- **Duration:** ~2-3 weeks.

- **Dependencies:** Planning completion.

3. Design Phase

- **Task:** Create architectural designs, database schemas, and user interface (UI) mockups.

- **Duration:** ~3-4 weeks.

- **Dependencies:** Completion of requirements.

4. Development (Coding) Phase

- **Task:** Build the software modules (e.g., room management, student registration, billing, reporting).

- **Duration:** ~6-10 weeks.

- **Dependencies:** Design completion.

5. Testing Phase

- **Task:** Conduct unit testing, integration testing, system testing, and user acceptance testing.

- **Duration:** ~3-5 weeks.

- **Dependencies:** Development completion.

6. Deployment Phase

- **Task:** Deploy the software to the hostel's servers or cloud environment.

- **Duration:** ~1 week.

- **Dependencies:** Successful testing.

7. Maintenance Phase

- **Task:** Provide ongoing support, bug fixes, and updates.

- **Duration:** Ongoing.

- **Dependencies:** Deployment completion.

